

Industrial Security (Master)

Solution Sheet – Section 05 Secure-by-Design Architectures

Exercise 5.1 – Mark scheme

Pass requires named zones (e.g. Plant, Process Control, Safety, Site IT) each with a justifiable SLT (typically 3 for control, 4 for safety), per-conduit allowlists, and a defensible diode placement (typically safety→site for telemetry egress).

Exercise 5.2 – Mark scheme

Strong answers point to IEC 62443-4-1 evidence requirements (process maturity), CycloneDX/SPDX for SBOM, 90-day SLA with reasonable carve-outs, 10+ year support window, and signed downloads with vendor key rotation.

Exercise 5.3 – Mark scheme

Award marks for: explicit pass/fail per step, traceable artefacts (vendor signature, PCAP, vuln report), and a backup/restore that completes within RTO.

Exercise 5.4 – Mark scheme

Sensible sequencing (visibility before segmentation, segmentation before retirement), named owners, realistic cost bands, and risk reduction stated in process terms (downtime hours, safety events).